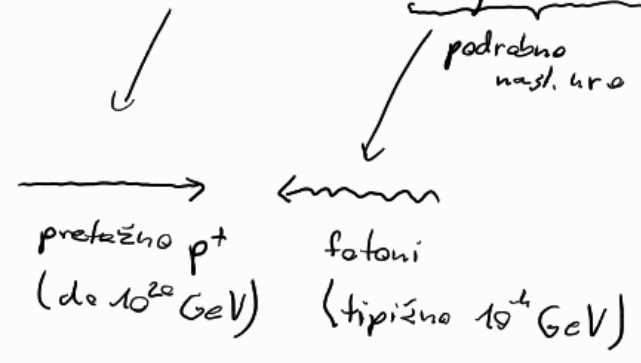


Zgled Produkcija parov e^+e^- ob sipanju kozmičnih žarkov na sevanju 3K ozadja

proces: $\gamma p \rightarrow p e^+ e^-$



LAB pred trkam:

proton: $p^\mu = (E, \vec{p}) = (E, p, 0, 0)$

foton: $p^\mu = (E_\gamma, -p_\gamma, 0, 0)$

TEŽ po trku:

četverec = $((m_p + 2m_e)c^2, 0)$

→ izračun invariante:

$$(E + E_\gamma)^2 - (p - p_\gamma)^2 = ((m_p + 2m_e)c^2)^2$$

$$\cancel{E^2} + 2EE_\gamma + \cancel{E_\gamma^2} - \cancel{p^2} + 2pp_\gamma - \underbrace{\cancel{p_\gamma^2}}_{E_\gamma^2} = \cancel{m_p^2 c^4} + 4m_p c^2 m_e + 4m_e^2 c^4$$

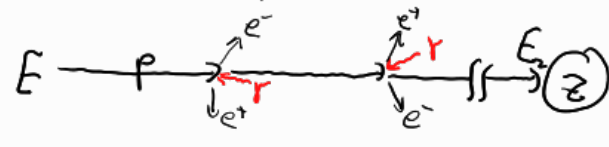
URL (ultra relat. limit): $m_p c^2 \approx 1\text{GeV} \rightarrow E \approx pc$ (zares $E = \sqrt{p^2 c^2 + m_p^2 c^4}$)

↑ ↑
aglomer majhen

$\cancel{4EE_\gamma} = \cancel{4m_p c^2 m_e c^2} + \cancel{4m_e^2 c^4} \rightarrow m_e \ll m_p$

$EE_\gamma = m_p c^2 m_e c^2$

→ $E = \frac{m_p m_e c^4}{E_\gamma} = 10^{16} \text{eV}$ → ad te E dalje je proces $p\gamma \rightarrow p e^+ e^-$ učinkovit pri izgubljanju energije kozmičnih žarkov

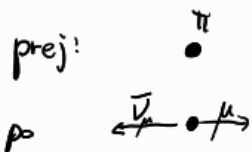


razpad π^- v mirovanju $\pi^- \rightarrow \mu^- \bar{\nu}_\mu$
 pion μ^- $\bar{\nu}_\mu$ $\leftarrow$ mionski anti-neutrino

$$m_\pi c^2 \approx 139,6 \text{ MeV}$$

$$m_\mu c^2 \approx 105,7 \text{ MeV}$$

$$m_\nu c^2 \approx 0 \text{ (} 10^{-5} \text{ eV)}$$



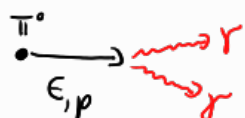
$$\begin{aligned} m_\pi^2 c^4 &= E_\mu + E_\nu \quad \text{si razdelita} \\ 0 &= p_\mu + p_\nu = p_\mu + \frac{E_\nu}{c} \rightarrow E_\nu^2 = (p_\mu c)^2 = E_\mu^2 - m_\mu^2 c^4 \\ (m_\pi^2 c^4 - E_\mu)^2 &= E_\nu^2 \end{aligned}$$

$$m_\pi^2 c^4 - 2m_\pi c^2 E_\mu + E_\mu^2 = E_\mu^2 - m_\mu^2 c^4$$

$$E_\mu = \frac{m_\pi^2 c^4 + m_\mu^2 c^4}{2m_\pi c^2} \rightarrow T_\mu = E_\mu - m_\mu c^2 = 4,1 \text{ MeV}$$

} skupaj $m_\pi - m_\mu$
 in še $E_\nu = 29,8 \text{ MeV}$

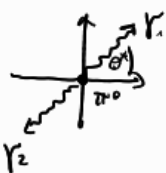
razpad v letu: $\pi^0 \rightarrow 2\gamma$



$$m_\pi c^2 \approx 135 \text{ MeV}$$

$$E_\pi = 6 \text{ GeV (podatek nealuge)}$$

Tec:

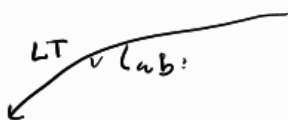


$$p_{1x}^* c = -p_{2x}^* c = \frac{m_\pi c^2}{2} \cdot \cos \theta^*$$

$$p_{1y}^* c = -p_{2y}^* c = \frac{m_\pi c^2}{2} \cdot \sin \theta^*$$

$$\gamma^* = \frac{1}{\sqrt{1-\beta^{*2}}} = \frac{E_\pi}{\sqrt{E_\pi^2 + p_\pi^2 c^2}} = \frac{E_\pi}{m_\pi c^2} \approx 46 = \gamma$$

$\uparrow$
 $= p_\pi c / E_\pi$



$$p_{1x} = \gamma \left(p_{1x}^* + \beta \frac{m_\pi c}{2} \right) = \frac{m_\pi c}{2} \left(\gamma \cos \theta^* + \sqrt{\gamma^2 - 1} \right)$$

$$p_{1y} = -p_{2y} = \frac{m_\pi c}{2} \sin \theta^*$$

$$p_{2x} = \gamma \left(p_{2x}^* + \beta \frac{m_\pi c}{2} \right) = \frac{m_\pi c}{2} \left(-\gamma \cos \theta^* + \sqrt{\gamma^2 - 1} \right)$$

$$\tan \theta_1 = \frac{p_{1x}}{p_{1y}} = \frac{\sin \theta^*}{r \cos \theta^* + \sqrt{r^2 - 1}}$$

$$\tan \theta_2 = \frac{p_{2x}}{p_{2y}} = \frac{\sin \theta^*}{-r \cos \theta^* + \sqrt{r^2 - 1}}$$

$$\rightarrow \tan \theta = \tan(\theta_1 - \theta_2) = \frac{\tan \theta_1 - \tan \theta_2}{1 + \tan \theta_1 \tan \theta_2}$$

$$\text{v primeru } r \gg 1: \tan \theta = \frac{2}{r \sin \theta^*}$$

Edino kadar je θ^* res majhen,
r odletita naprej